AHSIG-SSR: Auxiliary Hyperspectral Image-Guided Model for Spectral Super-Resolution

Cong Liu, Jinling Cai, and Faming Fang

***Abstract*—Recently, the spectral super-resolution (SSR) method has attracted increasing attention, which generates a hyperspectral image (HSI) by using a corresponding multispectral image (MSI). However, existing SSR methods usually inevitably have some limitations. Prior-driven SSR methods are difficult to obtain satisfactory results because of the severe lack of spectral information of the MSI. Spectral library-based SSR methods are affected by the incompleteness and poor generalizability of existing spectral libraries. In addition, deep learning-based SSR approaches, despite their impressive performance, demand large-scale paired datasets and often lack robustness across different sensors and scenes. To alleviate these limitations, we propose a novel SSR framework that uses an easily accessible HSI as auxiliary information to enhance the accuracy and convenience of the SSR method, which circumvents the need for accurate registration and extensive training data. Specifically, we design three models to capture the spectral information from this auxiliary HSI and take the learned spectral information into the spectral reconstruction of MSIs: 1) We design adaptive local spectral dictionary learning (ALSDL) to learn the spectral dictionary from the auxiliary HSI; 2) we design low-rank sparse spectral correlation modeling (LRS-SCM) to learn the spectral correlation from the auxiliary HSI; and 3) we design similar spectral fibers filling (SSFF) model to extract the similar spectral fibers from the auxiliary HSI. Experimental results on multiple hyperspectral datasets demonstrate the efficacy of the proposed method. The code is available at: https://github.com/6845698566/AHSIG-SSR**

***Index Terms*—Similar hyperspectral fibers, spectral correlation, spectral dictionary learning, spectral super-resolution (SSR).**

I. Introduction

HYPERSPECTRAL imaging has emerged as an advanced technology that captures abundant spectral signatures alongside detailed spatial information by acquiring numerous narrow and contiguous spectral bands. Compared with conventional RGB images, hyperspectral images (HSIs) possess a superior capability to reveal the intrinsic characteristics of diverse objects and materials because of their rich spectral information. Consequently, HSIs have found extensive applications in various fields, including remote sensing [1], [2], anomaly detection [3], [4], mineral exploration [5], [6], medical diagnostics [7], agricultural monitoring [8], and image classification [9], [10].

However, how to effectively acquire high-quality HSIs remains a fundamental challenge in this domain. A straightforward solution lies in the development of high-precision HSI sensors. However, it is well recognized that enhancing the spectral resolution inevitably compromises spatial resolution due to the limitations imposed by solar irradiance. In other words, it is extremely difficult to simultaneously acquire HSIs with high spectral and high spatial resolution [11], [12]. To address this inherent tradeoff, three dominant strategies have been proposed: spatial super-resolution (Spatial SR) [13], [14], HSI-MSI fusion [15], [16], and spectral super-resolution (SSR). Spatial SR focuses on reconstructing a high-spatial-resolution HSI (HR-HSI) from its corresponding low-spatial-resolution HSI (LR-HSI) captured by hyperspectral sensors [17]. HSI-MSI fusion techniques aim to generate an HR-HSI by combining an LR-HSI with a high-spatial-resolution multispectral image (HR-MSI) or an RGB image characterized by low spectral resolution [18], [19]. However, both approaches typically depend on expensive HSI sensors and are time-intensive due to the prerequisite of LR-HSI acquisition [20], [21]. Furthermore, the effectiveness of HSI-MSI fusion is contingent on precise image registration between the two modalities, which remains a significant challenge in real-world scenarios [22], [23]. In contrast, SSR methods circumvent these limitations by reconstructing HR-HSIs from HR-MSIs, which provide high spatial resolution but limited spectral information. Compared with acquiring LR-HSIs, obtaining HR-MSIs or RGB images is considerably more cost-effective and convenient. As a result, SSR offers a more practical and flexible solution for a wide range of real-world applications [24], [25], [26]. In recent years, a wide array of SSR methodologies have been introduced which can be broadly categorized into two classes: model-based approaches and deep learning-based approaches. Model-based SSR methods can be further subdivided into prior-driven techniques and spectral library-driven techniques. Prior-driven methods typically formulate SSR as an inverse problem by taking advantage of the intrinsic structural characteristics of

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the observed MSI, employing techniques such as sparse representation [27], dictionary learning [28], and low-rank modeling [29]. Although these methods have demonstrated promising SSR performance, several inherent limitations remain. First, reconstructing accurate spectral information from the spectrally limited MSI remains fundamentally challenging. Second, hand-crafted priors alone often fail to comprehensively capture the complex structural dependencies present in natural scenes. Third, because of the absence of training data requirements, these methods frequently struggle to maintain reconstruction accuracy when dealing with HSIs exhibiting intricate spatial and spectral variability. Alternatively, spectral library-driven approaches rely on the construction of an offline spectral library, which is subsequently used for HSI reconstruction by spectral matching, regression modeling, or sparse representation [30]. However, these methods are hampered by limited generalization capabilities, poor adaptability to diverse or dynamic environments, and the high cost associated with spectral library acquisition, thereby constraining their practical applicability.

More recently, deep learning techniques, renowned for their powerful nonlinear learning and representation capabilities, have gained significant traction across a wide range of HSI tasks. Deep learning-based SSR approaches aim to establish complex nonlinear mappings from MSI to HSI by leveraging the expressive power of neural networks. These methods have achieved outstanding reconstruction performance when trained on large-scale, high-quality paired datasets [31], with representative advancements including deep blind super-resolution networks [32], attention-enhanced architectures [33], and function-mixing models [34]. Nonetheless, practical deployment of these approaches is hindered by the scarcity of suitable training data, given that HSIs are typically acquired under varying sensor specifications and scene conditions. Furthermore, the substantial computational cost associated with training, coupled with complex model deployment pipelines, poses additional barriers to their real-world application.

A. Motivation and Contribution

Although the aforementioned approaches have demonstrated promising performance, all face significant limitations. HSI-MSI fusion methods are highly dependent on the precise image registration of the HSI and the MSI. Prior-driven SSR methods are usually hard to recover complete spectral information using only hand-crafted priors. Spectral library-based methods usually require an accurate spectral library. Deep learning models require extensive labeled data and tend to generalize poorly to new environments. These challenges become more severe in real-world settings characterized by sensor heterogeneity, scene complexity, and uncalibrated spectral responses. These existing limitations motivate us to explore some novel methods which not only increase the spectral reconstruction accuracy but also require less additional information.

With all of the above analyses in mind, in this article, we propose a novel auxiliary HSI guidance model for the SSR task (AHSIG-SSR). First, we apply an HSI as the auxiliary HSI which is obtained from the same HSI sensor as the desired HR-HSI, and thus it has the same spectral band number as the desired HR-HSI. Second, we design three strategies to fully capture the spectral information from this auxiliary HSI to better reconstruct the spectral information of the desired HSI. Specifically, the contributions of this work are summarized in the following.

1) We propose an adaptive local spectral dictionary learning (ALSDL) model, which adaptively learns multiple spectral dictionaries from the auxiliary HSI and utilizes these dictionaries to guide the spectral reconstruction process.
2) We design a low-rank sparse spectral correlation modeling (LRS-SCM), which captures the low-rank sparse spectral correlation among the spectral bands of the auxiliary HSI and applies this model to the spectral reconstruction.
3) We develop a similar spectral fibers filling (SSFF) model, which searches for a hyperspectral fiber similar to the desired HR-HSI from the auxiliary HSI and applies this hyperspectral fiber to fill the desired HR-HSI.
4) Our model requires no pretraining and achieves state-of-the-art performance across multiple hyperspectral datasets, effectively improving MSI spectral resolution while ensuring superior spectral reconstruction quality compared to existing methods.

The remainder of this article is organized as follows. Section II reviews related SSR methods, covering both model-based and deep learning-based techniques. Section III details the proposed AHSIG-SSR framework. Section IV details the optimization strategy. Section V presents extensive experiments and discussions on three hyperspectral datasets. Finally, Section VI concludes the article.

II. Related Work

This section will introduce some related work that can be categorized into model-based and deep learning-based methods.

A. Model-Based Methods

Model-based SSR methods typically formulate the reconstruction as an optimization problem with carefully designed spectral priors to constrain the solution space. These approaches are mainly divided into two subcategories.

1) Prior-Driven SSR Methods: These methods rely solely on the input MSI and attempt to extract spectral priors from the HSI itself. Arad and Ben-Shahar [35] developed a sparse representation method using a learned spectral–RGB dictionary for efficient hyperspectral recovery from RGB images. Jia et al. [36] proposed a manifold-based spectral reconstruction method that maps RGB images to hyperspectral signatures by learning a low-dimensional embedding of natural spectra using nonlinear dimensionality reduction. Wu et al. [37] adapted the A+ regression model for spectral reconstruction, achieving CNN-level precision with improved efficiency and lower data demand. Yi et al. [38] integrated spectral response modeling with dictionary learning for accurate spectral reconstruction, while enforcing spatial consistency through spatial constraints and low-rank priors.

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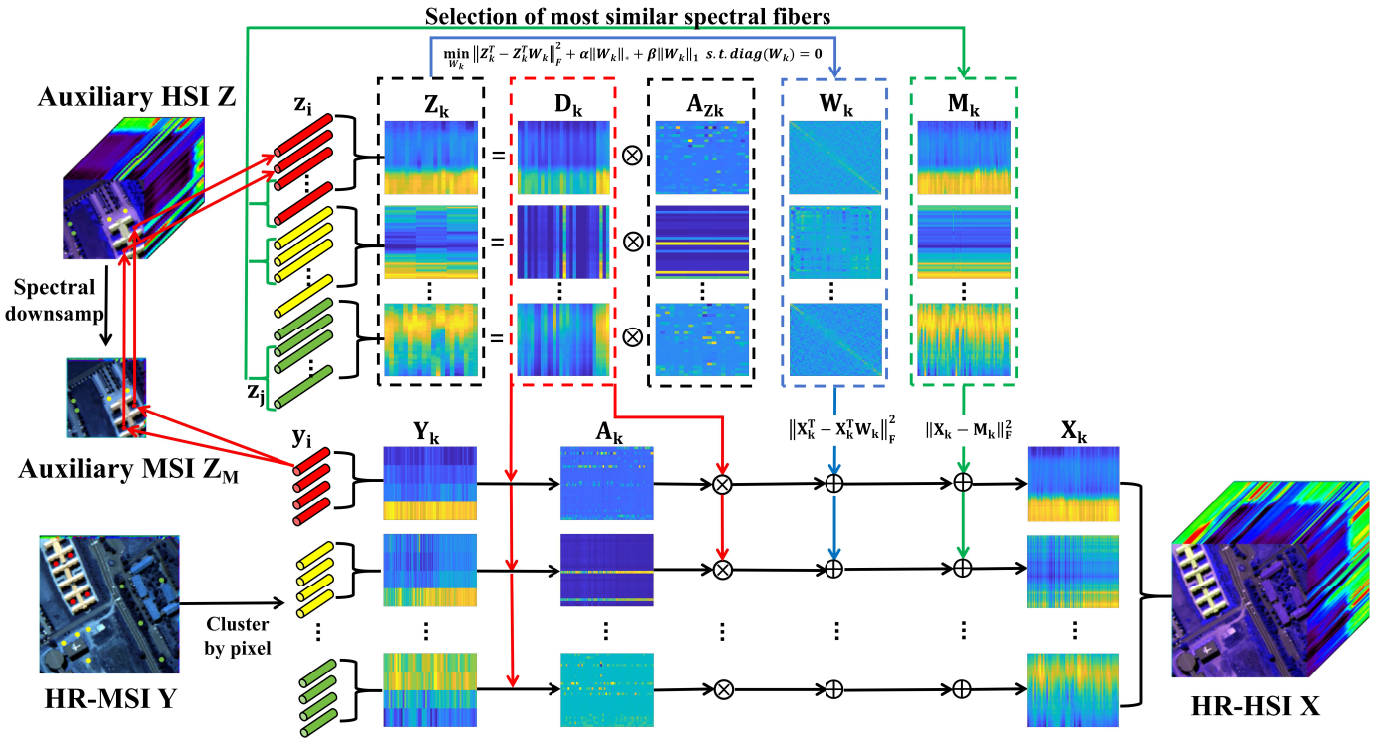


Fig. 1. Structure of the proposed AHSIG-SSR.

2) Spectral Library-Based Methods: These approaches exploit externally collected hyperspectral databases to guide reconstruction. Arad and Ben-Shahar [35] introduced a sparse representation framework for RGB-to-HSI mapping by retrieving the most similar spectral samples from a precompiled library. Sha et al. [39] proposed a full-spectrum SSR method based on the linear spectral mixing model (LSMM), where MSI is spectrally unmixed and reconstructed into full-spectrum HSI using spectral libraries and consistent abundance estimates. Yan et al. [40] proposed a dictionary optimization framework that adapts public spectral libraries to scene-specific domains, incorporating spatial and spectral cues during reconstruction. Although theoretically promising, these methods often suffer from limited generalizability, poor scene matching, and high acquisition costs associated with collecting high-quality spectral libraries.

B. Deep Learning-Based Methods

Deep learning-based SSR methods have evolved rapidly, moving from purely data-driven networks to more physically informed, efficient architectures. Early work by Galliani et al. [41] demonstrated that a very deep 56-layer CNN can learn a nonlinear mapping from RGB to hyperspectral space, while Shi et al. [42] and Martínez et al. [43] further improved reconstruction accuracy by integrating residual-dense connections and pretrained deep models, respectively. Recognizing that depth alone is not sufficient, Li et al. [44] introduced a dual-branch attention mechanism to adaptively fuse spectral features and preserve fine details. To reduce the dependency on labeled pairs, Qu et al. [45] proposed an unsupervised framework that simultaneously learns image priors and mapping functions. Based on these insights, Li et al. [46] incorporated camera spectral sensitivity as a physical prior within a hybrid 2-D–3-D CNN equipped with second-order channel and band attention, and He et al. [47] embedded the known spectral response function (SRF) into a variationally unfolded CNN with channel attention and spectral angle loss for enhanced interpretability. More recently, Dian et al. [48] developed a model-guided cross-fusion network that alternates between dual-branch prior learning and imaging model-driven optimization, producing state-of-the-art accuracy with a compact model size. Despite these advances, current deep SSR approaches still rely heavily on large paired datasets, struggle to generalize across different sensors and scenes, and often incur substantial computational overhead, highlighting the need for more robust, lightweight, and unsupervised solutions.

III. Proposed AHSIG-SSR Method

In this study, we will introduce the proposed AHSIG-SSR method, which contains ALSDL, LRS-SCM, and SSFF. All three modules are designed to capture the spectral information from the auxiliary HSI. The overall framework of the proposed SSR approach is illustrated in Fig. 1.

A. Problem Formulation

In the proposed method, $\mathbf{X} \in \mathbb{R}^{\lambda_H \times N}$ and $\mathbf{Y} \in \mathbb{R}^{\lambda_M \times N}$ denote the desired HR-HSI and the observed HR-MSI, respectively, where λ_M and λ_H denote the number of spectral bands in HR-MSI and HR-HSI, respectively. They are satisfied with

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($\lambda_M \ll \lambda_H$). N represents the number of pixels per band. The degradation model can be expressed as

$$\mathbf{Y} = \mathbf{F}\mathbf{X} + \mathbf{N}_\mathbf{Y} \tag{1}$$

where $\mathbf{F} \in \mathbb{R}^{\lambda_M \times \lambda_H}$ and $\mathbf{N}_\mathbf{Y}$ represent the SRF and the noise, respectively.

It is an ill-posed problem to directly reconstruct $\mathbf{X}$ from $\mathbf{Y}$ due to $\lambda_M \ll \lambda_H$. Therefore, most of the existing methods usually design different regularization terms to address this issue, which can be formulated as

$$\arg\min_{\mathbf{X}} \|\mathbf{Y} - \mathbf{F}\mathbf{X}\|_F^2 + \phi(\mathbf{X}) \tag{2}$$

where $\phi(\mathbf{X})$ denotes additional priors on $\mathbf{X}$. However, the spectral information of $\mathbf{Y}$ is seriously missing and it is difficult to reconstruct $\mathbf{X}$ only using $\phi(\mathbf{X})$. Therefore, we introduce an auxiliary HSI with complete spectral information to guide the spectral reconstruction, which can be formulated as

$$\arg\min_{\mathbf{X}} \|\mathbf{Y} - \mathbf{F}\mathbf{X}\|_F^2 + \phi(\mathbf{Z} \to \mathbf{X}). \tag{3}$$

$\mathbf{Z} \in \mathbb{R}^{\lambda_H \times N_Z}$ represents the auxiliary HSI, which has the same spectral bands λ_H as the desired HR-HSI $\mathbf{X}$. N_Z denotes the number of pixels per band. $\phi(\mathbf{Z} \to \mathbf{X})$ represents the guided information from $\mathbf{Z}$ to $\mathbf{X}$. It should be noted that no registration operation is required between the observed HR-MSI and the auxiliary HSI, which differs from the HSI-MSI fusion task.

B. Similar Hyperspectral Fibers Grouping and Matching

The observed HR-MSI $\mathbf{Y}$ contains various different scenes. For each scene, it is the best way to search for hyperspectral fibers from the auxiliary HSI $\mathbf{Z}$ similar to this scene to reconstruct the desired hyperspectral fibers. Therefore, we design a similar spectral fiber grouping and matching method to achieve this goal, which contains *hyperspectral fiber grouping* and *searching for similar hyperspectral fibers*.

1) Hyperspectral Fibers Grouping: $\mathbf{Y}$ is partitioned into several groups along the spatial direction using the K-means method, and each group corresponds to a scene. The kth group can be represented as $\mathbf{Y}_k \in \mathbb{R}^{\lambda_H \times N_k}$, where N_k is the number of spectral fibers in the kth group.

2) Searching for Similar Hyperspectral Fibers: For the kth group $\mathbf{Y}_k$, we search for G hyperspectral fibers denoted as $\mathbf{Z}_k \in \mathbb{R}^{\lambda_H \times G}$ that are most similar to it from the auxiliary HSI $\mathbf{Z}$. Specifically, we first transform $\mathbf{Z}$ into a downsampled auxiliary MSI $\mathbf{Z}_M \in \mathbb{R}^{\lambda_M \times N_Z}$ using the spectral downsampling matrix $\mathbf{F}$, to ensure the spectral alignment between $\mathbf{Y}_k$ and $\mathbf{Z}_M$. Then, we search for the top-g most similar hyperspectral fibers from $\mathbf{Z}_M$ for each hyperspectral fiber of $\mathbf{Y}_k$. Finally, all selected similar spectral fibers are aggregated to form the matrix $\mathbf{Z}_{M_k} \in \mathbb{R}^{\lambda_M \times G}$. We set $g = 2$ in the following experiments to reduce the computational burden. G denotes the total number of spectral fibers in $\mathbf{Z}_{M_k}$. Ideally, G is equal to $N_k \times g$. However, there are many hyperspectral fibers repeated when searching for different hyperspectral fibers of $\mathbf{Y}_k$. Hence, G is less than $N_k \times g$ in practice application. Since $\mathbf{Z}_M$ and $\mathbf{Z}$ share the same spatial structure, we can get $\mathbf{Z}_k$ by directly retrieving $\mathbf{Z}_M$.

Next, we will design three models to capture the spectral information of $\mathbf{Z}_k$ to assist the reconstruction of the desired HR-HSI $\mathbf{X}_k \in \mathbb{R}^{\lambda_H \times N_k}$.

C. Adaptive Local Spectral Dictionary Learning

Spectral dictionary learning is widely used in various HSI processing tasks, which can map the HSI from the original space to a low-dimensional space to capture more interesting information, allowing the HSI to be reconstructed within a low-dimensional subspace, thereby enhancing the stability and robustness of SSR. Based on sparse representation theory, we assume that $\mathbf{X}_k$ can be expressed using a spectral dictionary and a sparse coefficient matrix as follows:

$$\mathbf{X}_k = \mathbf{D}_k\mathbf{A}_k \tag{4}$$

where $\mathbf{D}_k \in \mathbb{R}^{\lambda_H \times L}$ represents the spectral dictionary, $\mathbf{A}_k \in \mathbb{R}^{L \times N_k}$ is the sparse coefficient matrix, and L denotes the number of dictionary atoms. Substituting this dictionary representation into (1) to obtain

$$\mathbf{Y}_k = \mathbf{F}\mathbf{D}_k\mathbf{A}_k + \mathbf{N}_\mathbf{Y}. \tag{5}$$

However, it is difficult to obtain an accurate spectral dictionary using only $\mathbf{Y}_k$ and (5). Since $\mathbf{Z}_k$ and $\mathbf{X}_k$ have the same spectral information, we can learn the spectral dictionary from $\mathbf{Z}_k$ and apply the learned spectral dictionary to the reconstruction of $\mathbf{Z}_k$, as can be seen from the red line in Fig. 1.

First, we learn the kth spectral dictionary $\mathbf{D}_k$ based on $\mathbf{Z}_k$ by optimizing the following objective function:

$$\begin{aligned} \min_{\mathbf{D}_k, \mathbf{A}_{Z_k}} \ & \left\|\mathbf{Z}_k - \mathbf{D}_k\mathbf{A}_{Z_k}\right\|_F^2 + \gamma\|\mathbf{D}_k\|_* + \eta\left\|\mathbf{A}_{Z_k}\right\|_1 \\ \text{s.t.} \ & \mathbf{D}_k \geq 0 \end{aligned} \tag{6}$$

where $\mathbf{A}_{Z_k} \in \mathbb{R}^{L \times G}$ is the sparse coefficient matrix. γ and η are regularization parameters. We impose the low-rank constraint on $\mathbf{D}_k$ to enforce dictionary compactness and enhance robustness against noise, which is a widely adopted strategy in signal processing [29]. Given the strong spectral correlations inherent in hyperspectral data, its principal information can be effectively represented using a limited set of basis vectors, reducing redundancy and improving generalization. In addition, the nonnegativity constraint on $\mathbf{D}_k$ ensures physically interpretable spectral signatures, further stabilizing the sparse representation and enhancing reconstruction accuracy.

Once the dictionary $\mathbf{D}_k$ is learned, we take the spectral dictionary into (3) to generate the following model:

$$\min_{\mathbf{A}_k, \mathbf{X}_k} \|\mathbf{Y}_k - \mathbf{F}\mathbf{D}_k\mathbf{A}_k\|_F^2 + \lambda_1\|\mathbf{A}_k\|_1 + \phi(\mathbf{Z}_k \to \mathbf{X}_k). \tag{7}$$

Through this process, ALSDL effectively leverages local spectral information of $\mathbf{Z}_k$ to enhance HSI super-resolution performance.

D. Low-Rank Sparse Spectral Correlation Modeling

Spectral correlation is an important characteristic of HSIs, representing the correlation between each spectral band and the other spectral bands. This is denoted as a spectral correlation matrix $\mathbf{W}_k \in \mathbb{R}^{\lambda_H \times \lambda_H}$ for the kth group, where the

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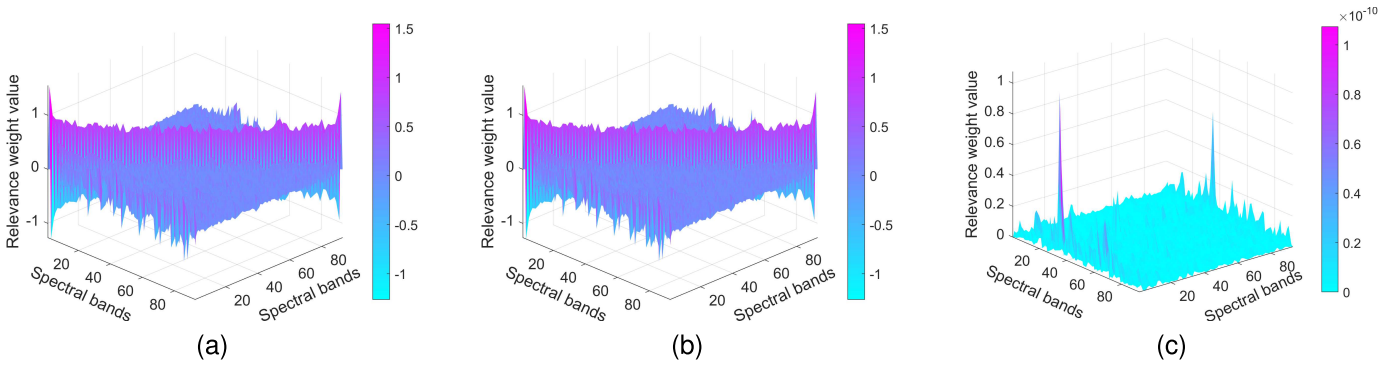


Fig. 2. (a) Spectral correlation matrix of $\mathbf{Z}_1$, (b) spectral correlation matrix of $\mathbf{X}_1$, and (c) difference between (a) and (b).

ith row of $\mathbf{W}_k$ represents the set of weights indicating the correlation between the ith spectral band and all other bands. Since $\mathbf{Z}_k$ and $\mathbf{X}_k$ have similar spectral information, their spectral correlation structures are also expected to be similar. Therefore, we propose to learn the spectral correlation from $\mathbf{Z}_k$ and transfer it to guide the reconstruction of $\mathbf{X}_k$. Fig. 2(a) and (b) shows the spectral correlation matrices of the first group $\mathbf{Z}_1$ and $\mathbf{X}_1$, respectively, on the Pavia University dataset (PaviaU). Fig. 2(c) visualizes the difference between these two matrices. As illustrated, the spectral correlation matrices of $\mathbf{Z}_1$ and $\mathbf{X}_1$ are highly similar, justifying our assumption. To ensure both local spectral smoothness and global consistency [49], we further incorporate low-rank and sparse regularization into the spectral correlation matrix to enhance its robustness. Moreover, we impose the constraint $\text{diag}(\mathbf{W}_k) = 0$, which prevents each spectral band from regressing onto itself. This avoids erroneous self-reinforcement and further stabilizes the reconstruction process.

Based on the above analyses, we first obtain the spectral correlation matrix $\mathbf{W}_k$ for the kth group of $\mathbf{Z}_k$ using the following function:

$$\begin{aligned} &\min_{\mathbf{W}_k} \ \left\|\mathbf{Z}_k^T - \mathbf{Z}_k^T\mathbf{W}_k\right\|_F^2 + \alpha\left\|\mathbf{W}_k\right\|_* + \beta\left\|\mathbf{W}_k\right\|_1 \\ &\text{s.t. } \text{diag}\left(\mathbf{W}_k\right) = 0 \end{aligned} \tag{8}$$

where α and β denote the regularization parameters for the low-rank and sparse constraints, respectively.

Then, we take $\mathbf{W}_k$ to establish the reconstruction module $\|\mathbf{X}_k^T - \mathbf{X}_k^T\mathbf{W}_k\|_F^2$ and take this module into (7) to obtain the following minimization:

$$\begin{aligned} \min_{\mathbf{A}_k,\mathbf{X}_k} \ &\|\mathbf{Y}_k - \mathbf{F}\mathbf{D}_k\mathbf{A}_k\|_F^2 + \lambda_1\left\|\mathbf{A}_k\right\|_1 \\ &+ \lambda_2\left\|\mathbf{X}_k^T - \mathbf{X}_k^T\mathbf{W}_k\right\|_F^2 + \phi\left(\mathbf{Z}_k \to \mathbf{X}_k\right) \\ \text{s.t. } &\mathbf{X}_k = \mathbf{D}_k\mathbf{A}_k \end{aligned} \tag{9}$$

where λ_2 is the regularization parameter.

E. Similar Spectral Fibers Filling

Similarity theory is another important characteristic widely used in various HSI processing tasks. The core of this theory is that there are many similar pixels in an image and the similar pixels can be reconstructed to each other. Inspired by similarity theory, we observed the spectral fibers of $\mathbf{X}$ and $\mathbf{Z}$, and found

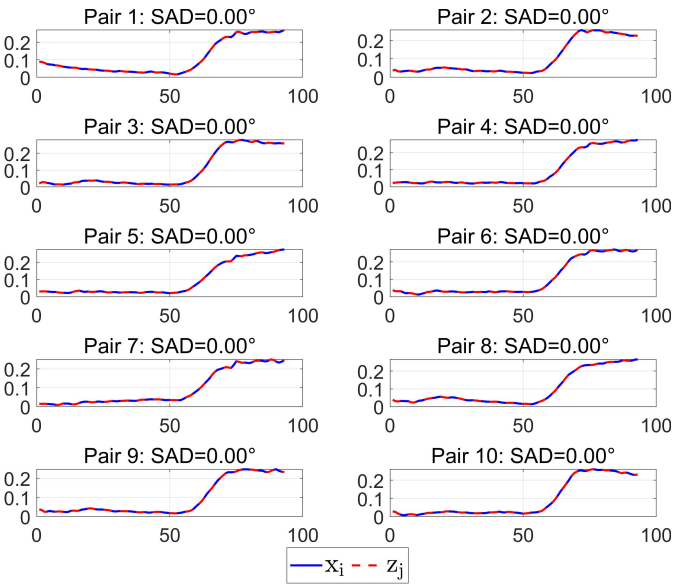


Fig. 3. Spectral curves of randomly selected ten hyperspectral fibers from $\mathbf{X}$ and their corresponding closest spectral fibers from $\mathbf{Z}$, respectively.

that almost each spectral fiber of $\mathbf{X}$ can find many similar spectral fibers in $\mathbf{Z}$. To confirm this idea, we randomly selected ten spectral fibers from $\mathbf{X}$ of the PaviaU and searched for the spectral fibers that are closest to them separately from $\mathbf{Z}$, as shown in Fig. 3. The spectral angle distance (SAD) is used to evaluate the similarity between them.

Based on the above analyses, we design a similar spectral fiber filling method. For each spectral fiber $\mathbf{x}_i$ in $\mathbf{X}$, we apply the most similar spectral fiber ($\mathbf{z}_j$) from $\mathbf{Z}$ to reconstruct it. However, it is hard to find a similar spectral fiber from the auxiliary HSI as $\mathbf{x}_i$ is unknown. It is fortunate that $\mathbf{y}_i$ is known, which can help us to find the most similar spectral fiber $\mathbf{z}_j$. When all similar spectral fibers are found, we can establish a similar HSI $\mathbf{M} \in \mathbb{R}^{\lambda_H \times N}$. The procedure of this process is summarized in Algorithm 1.

Next, we apply $\mathbf{M}$ to establish the reconstruction module $\|\mathbf{X} - \mathbf{M}\|_F^2$ and take this module into (9) to obtain the final optimization problem

$$\min_{\mathbf{A}_k,\mathbf{X}_k} \ \|\mathbf{Y}_k - \mathbf{F}\mathbf{D}_k\mathbf{A}_k\|_F^2 + \lambda_1\left\|\mathbf{A}_k\right\|_1$$

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Algorithm 1 Procedure of the SSFF

Input: Each observed HR-MSI group $\mathbf{Y}_k$, its corresponding similar downsampled auxiliary MSI group $\mathbf{Z}_{M_k}$, and the auxiliary HSI group $\mathbf{Z}_k$ that shares the same spatial domain with $\mathbf{Z}_{M_k}$
Output: the reconstructed similar HSI group image $\mathbf{M}_k$
Initialization: Set $i = 1$
while $i < N_k$ **do**
  Step 1: Taking out the ith spectral fiber $\mathbf{y}_i$ from $\mathbf{Y}_k$
  Step 2: Searching for the most similar spectral fiber $\mathbf{z}_{M_j}$ to $\mathbf{y}_i$ from $\mathbf{Z}_{M_k}$
  Step 3: Assign $\mathbf{m}_i \in \mathbf{M}_k$ as the corresponding fiber $\mathbf{z}_j \in \mathbf{Z}_k$, where $\mathbf{z}_j$ contains the full spectral information aligned with $\mathbf{z}_{M_j}$
  $i \leftarrow i + 1$
end while

$$+ \lambda_2 \left\| \mathbf{X}_k^T - \mathbf{X}_k^T \mathbf{W}_k \right\|_F^2 + \lambda_3 \left\| \mathbf{X}_k - \mathbf{M}_k \right\|_F^2$$
$$\text{s.t. } \mathbf{X}_k = \mathbf{D}_k \mathbf{A}_k \tag{10}$$

where $\mathbf{M}_k \in \mathbb{R}^{\lambda_H \times N_k}$ is the k group of $\mathbf{M}$, and λ_3 is the parameter that balances the contribution of $\mathbf{M}_k$.

IV. Model Optimization

Three models should be optimized, including the optimization of the spectral dictionary $\mathbf{D}_k$ in (6), the optimization of the spectral correlation matrix $\mathbf{W}_k$ in (8) and the overall model of (10). All three models are nonconvex optimization problems and will be solved using ADMM.

A. Dictionary Learning Optimization

For (6), we introduce an auxiliary variable $\mathbf{J}$ to impose a low-rank constraint on $\mathbf{D}_k$ and handle this constraint via a Lagrange multiplier $\boldsymbol{\lambda}$ along with a positive penalty parameter $\mu > 0$

$$\mathcal{L}\left(\mathbf{D}_k, \mathbf{A}_{Z_k}, \mathbf{J}, \boldsymbol{\lambda}\right) = \left\| \mathbf{Z}_k - \mathbf{D}_k \mathbf{A}_{Z_k} \right\|_F^2 + \gamma \|\mathbf{J}\|_* + \eta \|\mathbf{A}_{Z_k}\|_1 + \mu \left\| \mathbf{D}_k - \mathbf{J} + \frac{\boldsymbol{\lambda}}{2\mu} \right\|_F^2. \tag{11}$$

To solve (11), we adopt an alternating optimization strategy, iteratively updating each parameter while keeping the others fixed.

1) $\mathbf{A}_{Z_k}$ Subproblem:

$$\min_{\mathbf{A}_{Z_k}} \left\| \mathbf{Z}_k - \mathbf{D}_k \mathbf{A}_{Z_k} \right\|_F^2 + \eta \|\mathbf{A}_{Z_k}\|_1 \tag{12}$$

which can be solved by the soft-thresholding method

$$\mathbf{A}_{Z_k} = \text{Soft}_\eta\left(\left(\mathbf{D}_k^T \mathbf{D}_k \right)^{-1} \mathbf{D}_k^T \mathbf{Z}_k \right). \tag{13}$$

Here, $\text{Soft}_\eta(\cdot)$ denotes the element-wise soft-thresholding operator with threshold η, which is defined as: $\text{Soft}_\tau(x) = \max(x - \tau, 0)$.

Algorithm 2 Optimization Procedure for (11)

Input: $\mathbf{Z}_k$, parameters γ and η, tolerance ε, max iteration T_{max}.
Output: Spectral dictionary $\mathbf{D}_k$.
Initialization: Set $t = 0$; initialize $\mathbf{D}_k^{(0)}$, $\mathbf{J}^{(0)}$, $\mathbf{A}_{Z_k}^{(0)}$, $\boldsymbol{\lambda}^{(0)}$, and $\mu^{(0)}$.
while $t < T_{\max}$ **and** not converged **do**
  Step 1: Update $\mathbf{A}_{Z_k}^{(t)}$ using Eq. (13),
  Step 2: Update $\mathbf{D}_k^{(t)}$ using Eqs. (15) and (16),
  Step 3: Update $\mathbf{J}^{(t)}$ using Eq. (18),
  Step 4: Update the Lagrange multiplier using Eq. (19).
  $t \leftarrow t + 1$
end while

2) $\mathbf{D}_k$ Subproblem:

$$\min_{\mathbf{D}_k} \left\| \mathbf{Z}_k - \mathbf{D}_k \mathbf{A}_{Z_k} \right\|_F^2 + \mu \left\| \mathbf{D}_k - \mathbf{J} + \frac{\boldsymbol{\lambda}}{2\mu} \right\|_F^2. \tag{14}$$

Taking the derivative and setting it to zero, the analytical solution is

$$\mathbf{D}_k = \left(\mathbf{Z}_k \mathbf{A}_{Z_k}^T + \mu \mathbf{J} - \frac{\boldsymbol{\lambda}}{2} \right) \left(\mathbf{A}_{Z_k} \mathbf{A}_{Z_k}^T + \mu \mathbf{I} \right)^{-1}. \tag{15}$$

Then, project onto the nonnegative space

$$\mathbf{D}_k = \max\left(\mathbf{D}_k, 0 \right). \tag{16}$$

3) $\boldsymbol{J}$ Subproblem:

$$\min_{\mathbf{J}} \gamma \|\mathbf{J}\|_* + \mu \left\| \mathbf{D}_k - \mathbf{J} + \frac{\boldsymbol{\lambda}}{2\mu} \right\|_F^2 \tag{17}$$

which can be solved using singular value soft thresholding (SVT)

$$\mathbf{J} = \mathbf{U}\, \text{Soft}_{\frac{\gamma}{2\mu}}\left(\boldsymbol{\Sigma} \right) \mathbf{V}^T, \quad \text{where } \mathbf{D}_k + \frac{\boldsymbol{\lambda}}{2\mu} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^T. \tag{18}$$

4) Update Lagrange Multiplier $\boldsymbol{\lambda}$:

$$\boldsymbol{\lambda} = \boldsymbol{\lambda} + \mu \left(\mathbf{D}_k - \mathbf{J} \right). \tag{19}$$

The optimization procedure of (11) is summarized in Algorithm 2.

B. Spectral Correlation Matrix Optimization

For (8), we introduce two auxiliary variables $\mathbf{J}_1 = \mathbf{W}_k$ and $\mathbf{S}_1 = \mathbf{W}_k$. The augmented Lagrangian function is constructed as follows:

$$L\left(\mathbf{W}_k, \mathbf{J}_1, \mathbf{S}_1 \right) = \left\| \mathbf{Z}_k^T - \mathbf{Z}_k^T \mathbf{W}_k \right\|_F^2 + \alpha \|\mathbf{J}_1\|_* + \beta \|\mathbf{S}_1\|_1 + \mu_1 \left\| \mathbf{W}_k - \mathbf{J}_1 + \frac{\mathbf{Y}_1}{2\mu_1} \right\|_F^2 + \mu_1 \left\| \mathbf{W}_k - \mathbf{S}_1 + \frac{\mathbf{Y}_2}{2\mu_1} \right\|_F^2 \tag{20}$$

where $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are two Lagrange multipliers and $\mu_1 > 0$ is the positive penalty parameter. Equation (20) is also solved by using the alternating optimization strategy.

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1) $\mathbf{W}_k$ Subproblem:

$$\min_{\mathbf{W}_k} \left\|\mathbf{Z}_k^T - \mathbf{Z}_k^T\mathbf{W}_k\right\|_F^2 + \mu_1\left\|\mathbf{W}_k - \mathbf{J}_1 + \frac{\mathbf{Y}_1}{2\mu_1}\right\|_F^2 + \mu_1\left\|\mathbf{W}_k - \mathbf{S}_1 + \frac{\mathbf{Y}_2}{2\mu_1}\right\|_F^2. \tag{21}$$

This subproblem is a regularized least-squares optimization, whose closed-form solution can be derived as follows:

$$\mathbf{W}_k = \left(\mathbf{Z}_k^T\mathbf{Z}_k + 2\mu_1\mathbf{I}\right)^{-1}\left[\mathbf{Z}_k\mathbf{Z}_k^T + \mu_1\left(\mathbf{J}_1 - \frac{\mathbf{Y}_1}{2\mu_1} + \mathbf{S}_1 - \frac{\mathbf{Y}_2}{2\mu_1}\right)\right]. \tag{22}$$

To explicitly enforce the zero-diagonal constraint on $\mathbf{W}_k$, we update each diagonal element as

$$[\mathbf{W}_k]_{ij} = \begin{cases}[\mathbf{W}_k]_{ij}, & i \neq j\\ 0, & i = j\end{cases} \tag{23}$$

where $[\mathbf{W}_k]_{ij}$ denotes the element at the ith row and jth column of the weight matrix $\mathbf{W}_k$, where i and j represent the indices of the spectral vectors that are reconstructed and used for representation, respectively. This operation guarantees that no spectral vector is represented by itself, aligning with the nonself-representation assumption.

2) $\mathbf{J}_1$ Subproblem:

$$\min_{\mathbf{J}_1} \alpha\|\mathbf{J}_1\|_* + \mu_1\left\|\mathbf{W}_k - \mathbf{J}_1 + \frac{\mathbf{Y}_1}{2\mu_1}\right\|_F^2. \tag{24}$$

The solution can be obtained by applying SVT to the matrix

$$\mathbf{J}_1 = \mathbf{U}\,\mathrm{Soft}_{\frac{\alpha}{2\mu_1}}(\mathbf{\Sigma}_1)\,\mathbf{V}^T, \quad \text{where} \quad \mathbf{W}_k + \frac{\mathbf{Y}_1}{2\mu_1} = \mathbf{U}\mathbf{\Sigma}_1\mathbf{V}^T. \tag{25}$$

3) $\mathbf{S}_1$ Subproblem:

$$\min_{\mathbf{S}_1} \beta\|\mathbf{S}_1\|_1 + \mu_1\left\|\mathbf{W}_k - \mathbf{S}_1 + \frac{\mathbf{Y}_2}{2\mu_1}\right\|_F^2. \tag{26}$$

The sparse matrix $\mathbf{S}_1$ is updated by applying the soft-thresholding operator element-wise

$$\mathbf{S} = \mathrm{Soft}_{\frac{\beta}{2\mu_1}}\left(\mathbf{W}_k + \frac{\mathbf{Y}_2}{2\mu_1}\right). \tag{27}$$

4) Update Lagrange Multiplier $\mathbf{Y}_1$, $\mathbf{Y}_2$:

$$\mathbf{Y}_1 = \mathbf{Y}_1 + \mu_1(\mathbf{W}_k - \mathbf{J}_1), \quad \mathbf{Y}_2 = \mathbf{Y}_2 + \mu_1(\mathbf{W}_k - \mathbf{S}_1). \tag{28}$$

The optimization procedure of (20) is summarized in Algorithm 3.

C. Overall Model Optimization

To solve (10), we incorporate the constraint $\mathbf{X}_k = \mathbf{D}_k\mathbf{A}_k$ by introducing a Lagrange multiplier $\mathbf{\Lambda}$ and a positive penalty parameter $\rho > 0$

$$\mathcal{L}(\mathbf{A}_k, \mathbf{X}_k, \mathbf{\Lambda}) = \|\mathbf{Y}_k - \mathbf{F}\mathbf{D}_k\mathbf{A}_k\|_F^2 + \lambda_1\|\mathbf{A}_k\|_1 + \lambda_2\left\|\mathbf{X}_k^T - \mathbf{X}_k^T\mathbf{W}_k\right\|_F^2 + \lambda_3\|\mathbf{X}_k - \mathbf{M}_k\|_F^2 + \rho\left\|\mathbf{X}_k - \mathbf{D}_k\mathbf{A}_k + \frac{\mathbf{\Lambda}}{2\rho}\right\|_F^2. \tag{29}$$

We also employ an alternating minimization approach to optimize $\mathbf{A}_k$ and $\mathbf{X}_k$ separately.

Algorithm 3 Optimization Procedure for (20)

Input: The auxiliary HSI $\mathbf{Z}_k$, regularization parameters α and β, penalty parameter μ_1, tolerance ε, max iterations $T_{\max}$.
Output: Spectral correlation matrix $\mathbf{W}_k$.
Initialization: Set $t = 0$; initialize $\mathbf{W}_k^{(0)}$, $\mathbf{J}_1^{(0)}$, $\mathbf{S}_1^{(0)}$, $\mathbf{Y}_1^{(0)}$, and $\mathbf{Y}_2^{(0)}$.
while $t < T_{\max}$ **and** not converged **do**
  Step 1: Update $\mathbf{W}_k^{(t+1)}$ using Eqs. (22) and (23),
  Step 2: Update $\mathbf{J}_1^{(t+1)}$ using Eq. (25),
  Step 3: Update $\mathbf{S}_1^{(t+1)}$ using Eq. (27),
  Step 4: Update multipliers using Eq. (28).
  $t \leftarrow t + 1$
end while

1) $\mathbf{A}_k$ Subproblem:

$$\min_{\mathbf{A}_k} \|\mathbf{Y}_k - \mathbf{F}\mathbf{D}_k\mathbf{A}_k\|_F^2 + \lambda_1\|\mathbf{A}_k\|_1 + \rho\left\|\mathbf{X}_k - \mathbf{D}_k\mathbf{A}_k + \frac{\mathbf{\Lambda}}{2\rho}\right\|_F^2. \tag{30}$$

Since this problem involves an ℓ_1 constraint, we apply soft-thresholding for the solution. First, we compute the unconstrained solution

$$\mathbf{A}_k^{\frac{1}{2}} = \left(\mathbf{D}_k^T\mathbf{F}^T\mathbf{F}\mathbf{D}_i + \rho\mathbf{D}_k^T\mathbf{D}_k\right)^{-1}\left(\mathbf{D}_k^T\mathbf{F}^T\mathbf{Y}_k + \rho\mathbf{D}_k^T\mathbf{X}_k + \mathbf{D}_k^T\mathbf{\Lambda}\right). \tag{31}$$

Then, we apply soft-thresholding to account for the ℓ_1 constraint with λ_1

$$\mathbf{A}_k = \mathrm{Soft}_{\lambda_1}\left(\mathbf{A}_k^{\frac{1}{2}}\right). \tag{32}$$

2) $\mathbf{X}_k$ Subproblem:

$$\min_{\mathbf{X}_k} \lambda_2\left\|\mathbf{X}_k^T - \mathbf{X}_k^T\mathbf{W}_k\right\|_F^2 + \lambda_3\|\mathbf{X}_k - \mathbf{M}_k\|_F^2 + \rho\left\|\mathbf{X}_k - \mathbf{D}_k\mathbf{A}_k + \frac{\mathbf{\Lambda}}{2\rho}\right\|_F^2. \tag{33}$$

This is a standard quadratic optimization problem, whose closed-form solution is

$$\mathbf{X}_k = \left((\lambda_2 + \lambda_3 + \rho)\mathbf{I} - \lambda_2\left(\mathbf{W}_k + \mathbf{W}_k^T\right) + \lambda_2\mathbf{W}_k\mathbf{W}_k^T\right)^{-1} \times (\lambda_3\mathbf{M}_k + \rho\mathbf{D}_k\mathbf{A}_k - \mathbf{\Lambda}). \tag{34}$$

3) Updating the Lagrange Multiplier $\mathbf{\Lambda}$:

$$\mathbf{\Lambda} = \mathbf{\Lambda} + \rho(\mathbf{X}_k - \mathbf{D}_k\mathbf{A}_k). \tag{35}$$

The optimization procedure of (29) is summarized in Algorithm 4.

V. Experimental Results and Discussion

To evaluate the effectiveness of the proposed AHSIG-SSR method, we conduct extensive comparisons against nine representative approaches, covering two dominant paradigms in SSR research: *model-based methods* and *deep learning-based methods* to ensure a rigorous and balanced comparison. Model-based methods contain J-SLoL [29], DTS [28], UPSL_SSR [30], SSIC [50], and SLA_DOL [40]. Deep learning-based methods contain MSDN [51], SSRAN [52], LTRN [53], and

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Algorithm 4 Optimization Procedure for (29)

Input: Observed HSI $\mathbf{Y}_k$, dictionary $\mathbf{D}_k$, similar HSI $\mathbf{M}_k$, spectral correlation matrix $\mathbf{W}_k$; parameters $\lambda_1, \lambda_2, \lambda_3$, penalty ρ, maximum penalty $\rho_{\max}$, tolerance ϵ, max iterations $T_{\max}$.
Output: $\mathbf{X}_k$
Initialization: Set $t = 0$; initialize $\mathbf{X}_k^{(0)}$, $\mathbf{A}_k^{(0)}$, and $\mathbf{\Lambda}^{(0)}$.
while $t < T_{\max}$ **and** not converged **do**
    Step 1: Update $\mathbf{A}_k^{(t+1)}$ by using Eqs. (31) and (32),
    Step 2: Update $\mathbf{X}_k^{(t+1)}$ using Eq. (34),
    Step 3: Update Lagrange multiplier using Eq. (35).
    $t \leftarrow t + 1$
end while

SCDM [54]. This diverse selection of methods ensures that our evaluation robustly reflects the state-of-the-art in both traditional and data-driven SSR techniques. All methods were implemented using the parameter configurations reported in their original publications to ensure a fair and consistent comparison.

We adopt five metrics, including RMSE, PSNR, SAD, SSIM, and ERGAS, to evaluate SSR performance, which are widely used in many literature such as [29]. RMSE reflects spectral reconstruction accuracy, while SAD captures spectral distortion. PSNR and SSIM assess spatial fidelity and structural similarity, respectively. ERGAS provides a global measure of reconstruction quality. Lower RMSE, SAD, and ERGAS, along with higher PSNR and SSIM, indicate better performance.

To further validate the effectiveness of our approach, we performed qualitative comparisons by visual inspection of the reconstructed HSIs. In addition, we analyzed selected spectral fiber reconstructions to verify the spectral fidelity of the proposed approach. Experimental results show that our approach not only achieves superior spectral reconstruction accuracy but also effectively preserves spatial details, surpassing state-of-the-art techniques in both quantitative metrics and visual quality.

A. Datasets and Implementation Details

1) Pavia University Dataset: The PaviaU, acquired by the ROSIS-3 sensor over Pavia, Italy, comprises 610 × 340 pixels with 103 spectral bands (93 retained), and includes 42 776 labeled samples spanning nine land-cover classes. To simulate the SSR task, a predefined sensor response function was applied to the original high-resolution HSI to generate a $200 \times 200 \times 4$ HR-MSI patch (rows 151–350, columns 1–200). Subsequently, an auxiliary HSI was obtained by cropping a spatially shifted subcube of size 100 × 100 × 93 from the same HR-HSI volume (offset by 50 pixels), thus preserving the complete spectral information. For comparative evaluation, J-SLoL was supplied with an MSI extracted from rows 51–350 and cols 1–200, together with a $100 \times 200 \times 93$ HSI subregion cropped from the left portion of the scene. Library-based approaches (UPSL_SSR, SSIC, and SLA_DOL) replaced their spectral dictionaries with the auxiliary HSI, where UPSL_SSR additionally incorporated the region spanning rows 201–300 and cols 101–200. DTS was aligned accordingly, while deep learning models (MSDN, SSRAN, LTRN, and SCDM) were trained on the auxiliary HSI and evaluated on the MSI. For MSDN and SCDM, the four MSI bands were projected onto three RGB channels.

2) Indian Pines Dataset: The Indian Pines scene, acquired by AVIRIS over agricultural land in Indiana, comprises 145 × 145 pixels and 220 bands. We augment spectral priors with 518 USGS library spectra. For SSR, a $100 \times 100 \times 10$ MSI patch is synthesized using Sentinel-2 SRFs from a specific region (rows 46–145, cols 46–145) of the original HR-HSI. The auxiliary HSI ($50 \times 50 \times 220$) is directly cropped from a spatially shifted subregion (offset by 45 pixels) of the same HR-HSI, ensuring full spectral content. The comparative experimental settings follow prior work: J-SLoL uses the full MSI with an auxiliary HSI from the top-left 45 × 45 × 220 region; library-based methods (UPSL_SSR, SSIC, and SLA_DOL) substitute spectral dictionaries with our auxiliary HSI (UPSL_SSR uses two auxiliary cubes: rows 46–95 and 96–145); DTS aligns directly; and deep methods train on the auxiliary HSI and test on the MSI. MSDN and SCDM reduce the ten MSI bands to three channels.

3) CAVE Dataset: The CAVE dataset includes 32 high-resolution indoor scenes of size 512 × 512 with 31 spectral bands (400–700 nm at 10 nm intervals), offering high spatial and spectral fidelity. In our SSR setting, we construct the MSI by applying RGB response functions to the bottom half of each HR-HSI scene (256 × 512 × 3). The auxiliary HSI (256 × 512 × 31) is directly cropped from the top half of the same HR-HSI image, introducing natural spatial misalignment but preserving complete spectral information. The comparative experiments follow accordingly: J-SLoL uses the MSI and HSI halves as input; library-based approaches (UPSL_SSR, SSIC, and SLA_DOL) adopt our auxiliary HSI as their spectral dictionary (UPSL_SSR uses an additional region from the scene center); DTS uses the same split; and deep models are trained on the auxiliary HSI and evaluated on the MSI.

4) Chikusei Dataset: The Chikusei airborne scene comprises 128 spectral bands (0.4–1.0 μm) at a 2.5-m resolution, covering diverse urban, vegetation, and agricultural landscapes. To simulate SSR, we downsample the original 128-band HSI using a sensor response matrix to generate a 3-band HR-MSI. A 100×100 MSI target patch is selected, and the auxiliary HSI ($100 \times 100 \times 128$) is directly cropped from a spatially shifted region of the same HR-HSI, thus retaining full spectral information. The comparative experiments are adapted accordingly: J-SLoL uses the same MSI with an adjacent auxiliary HSI; UPSL_SSR, SSIC, and SLA_DOL utilize our auxiliary HSI as a spectral dictionary (UPSL_SSR uses additional regions from rows 21–120 and cols 21–120); DTS aligns directly; and deep SSR models train on the auxiliary HSI and test on the MSI.

B. Results and Analysis

1) Results on the PaviaU: Table I presents the quantitative comparison of various SSR methods on the PaviaU. The results indicate that traditional model-based techniques, such as J-SLoL, DTS, UPSL_SSR, SSIC, and SLA_DOL,

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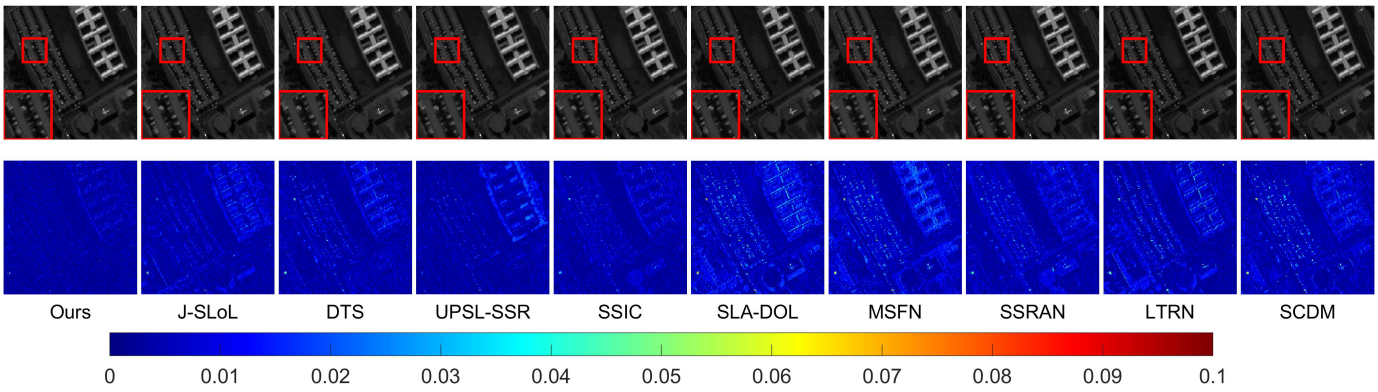


Fig. 4. Visual comparison of the 40th spectral band of HSIs reconstructed using different methods on the PaviaU. The first row shows the zoomed-in view reconstructions and the second row shows the corresponding pseudo-colored residual maps.

TABLE I

QUANTITATIVE COMPARISON ON THE PAVIAU

Method	PSNR	SAD	RMSE	SSIM	ERGAS
J-SLoL [29]	39.6449	0.0776	0.0145	0.9775	3.079
DTS [28]	42.5135	0.0386	0.0085	0.9870	1.9252
UPSL_SSR [30]	41.0064	0.0554	0.0109	0.9825	2.5149
SSIC [50]	41.6011	0.0401	0.0101	0.9866	2.2642
SLA_DOL [40]	40.1495	0.0730	0.0124	0.9796	2.9929
MSFN [51]	40.2865	0.0514	0.0119	0.9802	2.5374
SSRAN [52]	40.1957	0.0678	0.0122	0.9851	2.794
LTRN [53]	41.9591	0.0469	0.0097	0.9904	1.9937
SCDM [54]	41.8842	0.0404	0.0089	0.9852	2.0487
Ours	**46.2673**	**0.0263**	**0.0056**	**0.9943**	**1.2527**
Ideal Value	$\uparrow \infty$	$\downarrow 0$	$\downarrow 0$	$\uparrow 1$	$\downarrow 0$

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Method	PSNR	SAD	RMSE	SSIM	ERGAS
J-SLoL [29]	33.5709	0.0234	0.0132	0.6795	2.0842
DTS [28]	33.5166	0.0704	0.0155	0.7120	2.0900
UPSL_SSR [30]	33.3636	0.0204	0.0126	**0.7940**	1.9763
SSIC [50]	32.3051	0.0289	0.0135	0.7550	2.2164
SLA_DOL [40]	33.4303	0.0239	0.0126	0.7812	1.9830
MSFN [51]	33.9784	0.0221	0.0134	0.7789	2.0050
SSRAN [52]	32.8562	0.0297	0.0151	0.7420	2.3890
LTRN [53]	33.7628	0.0224	0.0118	0.7935	1.9140
SCDM [54]	34.0189	0.0192	0.0124	0.7707	1.9232
Ours	**39.2618**	**0.0118**	**0.0049**	0.7717	**0.8674**
Ideal Value	$\uparrow \infty$	$\downarrow 0$	$\downarrow 0$	$\uparrow 1$	$\downarrow 0$

capitalize on prior spectral knowledge to capture spectral correlations effectively. For example, J-SLoL, which exploits joint low-rank dictionary learning, demonstrates robust spectral structure preservation, whereas DTS benefits from a dictionary migration mechanism that enhances PSNR and reduces ERGAS, reflecting its generalization capability. Meanwhile, both UPSL_SSR and SSIC achieve competitive SSIM values by integrating spectral libraries with low-rank constraints, yet they still exhibit relatively higher SAD and RMSE, pointing to limitations in detailed spectral reconstruction. SLA_DOL, employing latent subspace transformation with adaptive dictionary learning, delivers moderate performance, suggesting residual issues in spectral alignment. In contrast, deep learning-based methods, namely, MSDN, SSRAN, LTRN, and SCDM, employ end-to-end spatial–spectral modeling to achieve a more balanced reconstruction quality. Although each exhibits strengths (with MSDN extracting multiscale features, SSRAN leveraging residual attention, LTRN modeling long-range dependencies via tensor decomposition, and SCDM employing a cascaded diffusion model to progressively refine spectral details and enhance reconstruction fidelity through stepwise denoising), they are still hindered by challenges in accurately restoring fine spectral details or capturing comprehensive spectral dependencies. Notably, our proposed method consistently outperforms all competing approaches across all evaluation metrics. With a significantly higher PSNR, markedly lower RMSE and SAD, and a near-ideal SSIM, our method demonstrates superior spectral reconstruction fidelity and exceptional structural consistency, affirming its efficacy and robustness in unregistered HSI-MSI SSR scenarios.

Fig. 4 presents the visual results for the 40th spectral bands of the PaviaU. The first row displays the magnified reconstructions obtained by ten competing methods, while the second row illustrates the corresponding pseudo-color residual maps. Visually, the proposed method demonstrates superior capability in preserving fine spatial and spectral structures, exhibiting sharper edges and fewer artifacts. In contrast, the other methods suffer from noticeable distortions and larger residuals, indicating inferior spectral reconstruction accuracy.

Fig. 5(a) compares the mean spectral curves averaged over a 5×5 spatial region in the PaviaU. The spectral curve reconstructed by our method is closely aligned with the ground truth, demonstrating superior accuracy in preserving spectral characteristics. In contrast, the curves obtained from other methods exhibit noticeable deviations, thereby underscoring the effectiveness of our approach in maintaining spectral fidelity.

2) Results on the Indian Pines Dataset: Table II summarizes the quantitative performance of various SSR methods on the Indian Pines dataset. The results reveal that traditional methods exhibit diverse strengths and limitations. For instance, J-SLoL demonstrates high performance in preserving spectral structure, as evidenced by favorable PSNR and SSIM values, yet its reconstruction accuracy is hampered by relatively elevated spectral angle (SAD) and RMSE. Similarly, while

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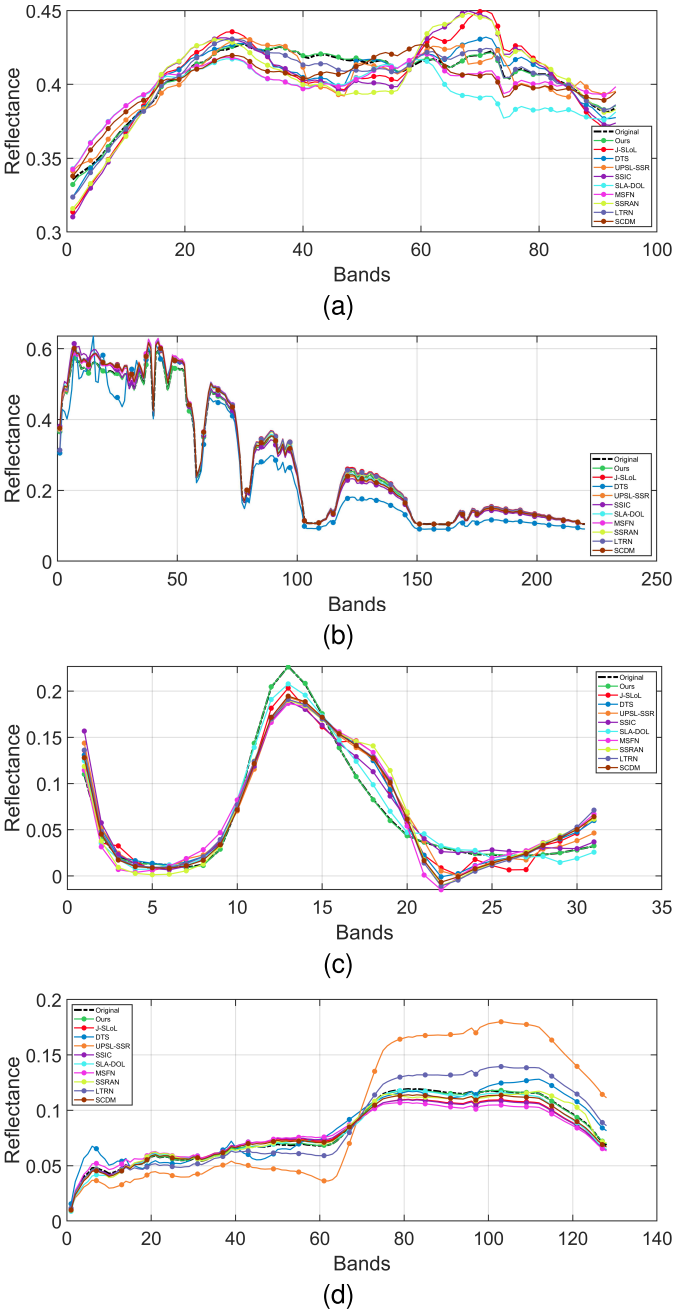


Fig. 5. Spectral signature comparison across four benchmark datasets. Reconstructed HSIs from various methods are evaluated by averaging the spectral responses over a 5×5 spatial region. The resulting curves represent the ability of each method to preserve the underlying spectral fidelity. (a) Pavia University. (b) Indian Pines. (c) CAVE–Superballs. (d) Chikusei.

DTS shows some improvement in reducing spectral angular deviation, its limited capability in modeling low-resolution spectral information leads to suboptimal results. Both UPSL_SSR and SSIC, which rely on spectral library priors and low-rank constraints, achieve competitive performance in terms of spectral angle preservation; however, they still struggle to fully recover fine spectral details, as indicated by their higher RMSE values. SLA_DOL, incorporating latent domain alignment techniques, yields moderate performance, suggesting residual challenges in reconciling spectral misalignments. In contrast, deep learning approaches—exemplified by MSDN, SSRAN, LTRN, and SCDM, employ end-to-end spatial–spectral modeling to achieve balanced reconstructions. Nonetheless, each of these methods also exhibits certain deficiencies, particularly in terms of local feature representation and overall spectral fidelity. Our proposed method, on the other hand, outperforms all competing approaches in key metrics. It achieves a significantly higher PSNR and markedly lower RMSE, SAD, and ERGAS, while maintaining a competitive SSIM. These results substantiate its superior ability to recover detailed spectral structures and robustly preserve spatial consistency.

TABLE III
QUANTITATIVE RESULTS ON THE CAVE DATASET

Method	PSNR	SAD	RMSE	SSIM	ERGAS
J-SLoL [29]	32.4043	0.1511	0.0341	0.9064	7.4820
DTS [28]	33.3759	0.2857	0.0282	0.9004	8.2461
UPSL_SSR [30]	33.8249	0.1813	0.0253	0.9142	6.0723
SSIC [50]	34.6072	0.1616	0.0248	0.9134	6.1050
SLA_DOL [40]	35.2487	0.1470	0.0619	**0.9200**	4.9375
MSFN [51]	34.1075	0.1762	0.0241	0.9018	5.9834
SSRAN [52]	33.2951	0.1928	0.0267	0.9063	6.3145
LTRN [53]	34.2396	0.1697	0.0238	0.9081	5.7742
SCDM [54]	34.1664	0.1895	0.0267	0.8927	5.7627
Ours	**36.6628**	**0.0692**	**0.0206**	0.9154	**4.5042**
Ideal Value	$\uparrow \infty$	$\downarrow 0$	$\downarrow 0$	$\uparrow 1$	$\downarrow 0$

Fig. 6 provides a visual comparison of the reconstructed HSIs on the 10th spectral bands of the Indian Pines dataset. The first row shows the enlarged reconstructions from ten competing methods, while the second row shows the corresponding pseudo-color residual images. It is noteworthy that our method can produce reconstructions with clearer details and minimal residuals. Similarly, Fig. 5(b) presents the spectral curves averaged over a 5×5 region in the Indian Pines dataset. The spectral response reconstructed by our method nearly overlaps with the ground truth, indicating its strong capability to faithfully recover the underlying spectral signatures.

Overall, our method attains state-of-the-art results on the Indian Pines dataset, demonstrating its robustness and effectiveness in complex scenarios. By incorporating a spectral correlation weight matrix, our approach further enhances SSR accuracy, enabling more precise HSI reconstruction.

3) Results on the CAVE Dataset: Table III presents the quantitative performance of various SSR methods on the CAVE dataset. For clear comparison, the best and second-best results are highlighted in bold and in italics, respectively. As shown in the table, traditional dictionary learning-based approaches such as J-SLoL [29] and DTS [28] show limited performance. Although J-SLoL achieves an acceptable SSIM of 0.9064, it suffers from a relatively low PSNR (32.4043) and a high RMSE (0.0341). DTS performs even worse in terms of spectral fidelity, recording the highest SAD (0.2857) and a poor ERGAS (8.2461), suggesting its limited ability to preserve spectral consistency. Spectral library-driven methods such as UPSL_SSR [30] and SSIC [50] offer improvements in RMSE and SAD, with UPSL_SSR achieving RMSE = 0.0253 and SAD = 0.1813. However, their ERGAS and SSIM scores still lag behind. SLA_DOL [40] reaches the

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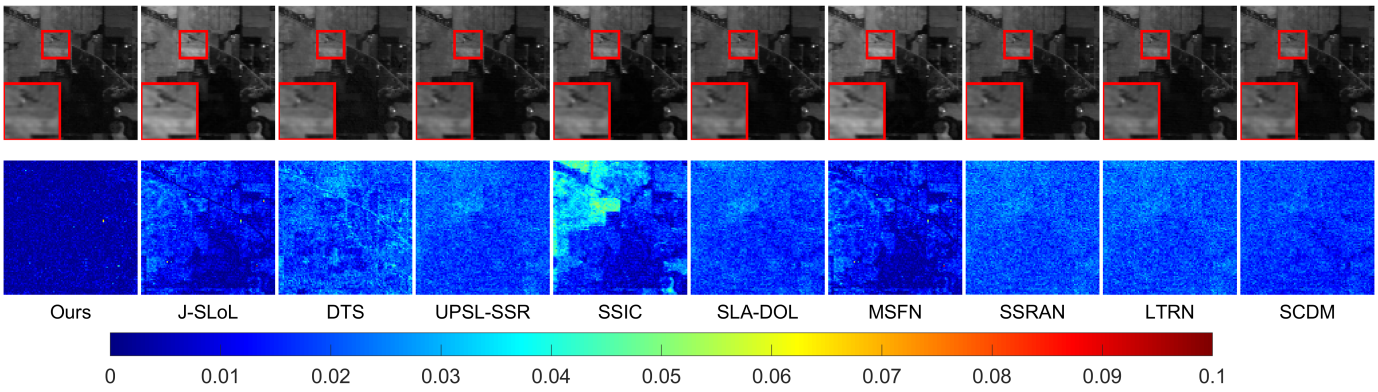


Fig. 6. Visual comparison of the 10th spectral bands of the HSIs reconstructed using different methods on the Indian Pines dataset. The first row shows the zoomed-in view reconstructions and the second row shows the corresponding pseudo-color residual maps.

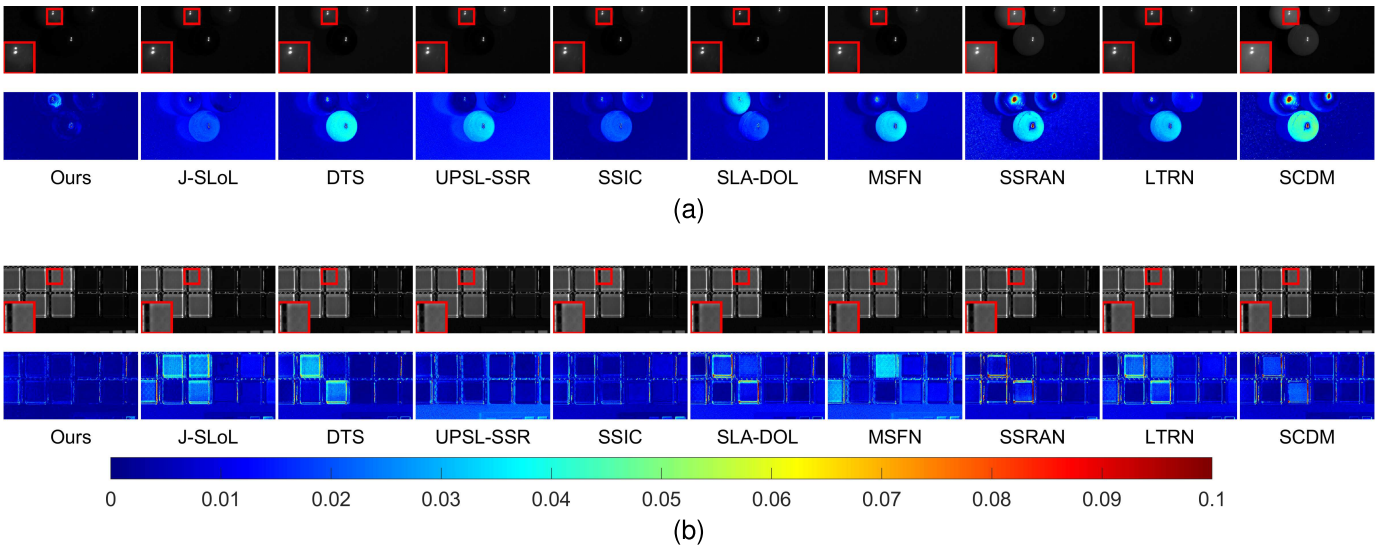


Fig. 7. Visual comparison of the 9th band of HSIs reconstructed from the CAVE dataset using different methods: (a) Superballs and (b) Glass_tiles. The first row shows the zoomed-in view reconstructions, and the second row presents the corresponding pseudo-color residual maps.

highest SSIM of 0.9200, but its notably high RMSE (0.0619) indicates excessive spectral smoothing or potential overfitting. Meanwhile, deep learning-based models, including MSFN [51], SSRAN [52], LTRN [53], and SCDM [54], demonstrate more balanced outcomes in most metrics. LTRN, for example, achieves a PSNR of 34.2396 and RMSE of 0.0238. However, their ERGAS values (all above 5) and SAD scores imply that there is room for improvement in the preservation of spectral detail. In contrast, our proposed method achieves the best performance in almost all evaluation criteria, with the highest PSNR (36.6628), the lowest SAD (0.0692), RMSE (0.0206), and ERGAS (4.5042), and a competitive SSIM (0.9154). These results demonstrate the effectiveness and robustness of our method for HSI super-resolution, especially under unregistered settings where traditional registration-based techniques tend to fail.

Fig. 7 presents the visual comparison of the 9th spectral band from two representative CAVE images, Superballs and Glass_tiles. For each image, the top row shows the zoomed-in view reconstructions obtained by nine competing methods, while the bottom row depicts the corresponding pseudo-color residual maps. In addition, Fig. 5(c) shows the spectral curves computed over a 5×5 region in the CAVE–Superballs image. Our method achieves an almost perfect match with the ground truth curve, further validating its robustness and precision in reconstructing fine spectral details. In contrast, other methods produce curves with more pronounced discrepancies.

TABLE IV
QUANTITATIVE COMPARISON ON THE CHIKESEI UNIVERSITY DATASET

Method	PSNR	SAD	RMSE	SSIM	ERGAS
J-SLoL [29]	26.5213	0.2807	0.0364	0.7123	12.4448
DTS [28]	27.1492	0.2905	0.0344	0.7641	11.9733
UPSL_SSR [30]	25.7182	0.3824	0.0447	0.8128	14.9536
SSIC [50]	27.4976	0.2642	0.0347	0.7339	11.7416
SLA_DOL [40]	28.1281	0.2680	0.0317	0.7952	10.7634
MSFN [51]	27.6203	0.2581	0.0341	0.7700	11.5292
SSRAN [52]	27.7434	0.2858	0.0328	0.7674	11.1879
LTRN [53]	28.2493	0.2894	0.0325	0.7732	10.9201
SCDM [54]	28.7800	0.1991	0.0284	0.7669	9.6697
Ours	**32.0589**	**0.1357**	**0.0199**	**0.8238**	**6.7582**
Ideal Value	$\uparrow \infty$	$\downarrow 0$	$\downarrow 0$	$\uparrow 1$	$\downarrow 0$

4) Results on the Chikusei Dataset: As shown in Table IV, our proposed method achieves the best performance across all evaluation metrics on the Chikusei dataset. Specifically,

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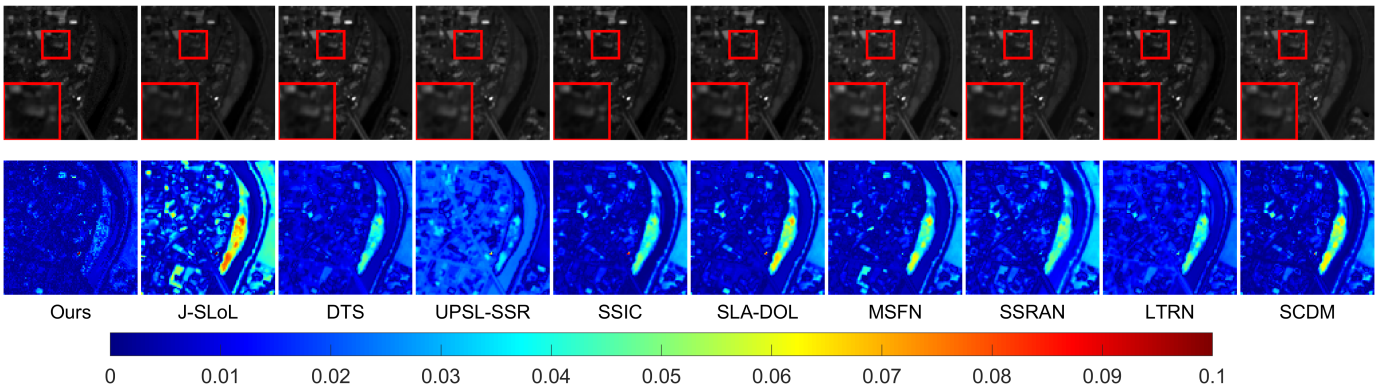


Fig. 8. Visual comparison of the 25th spectral bands of the HSIs reconstructed using different methods on the Chikusei dataset. The first row shows the zoomed-in view reconstructions and the second row shows the corresponding pseudo-color residual maps.

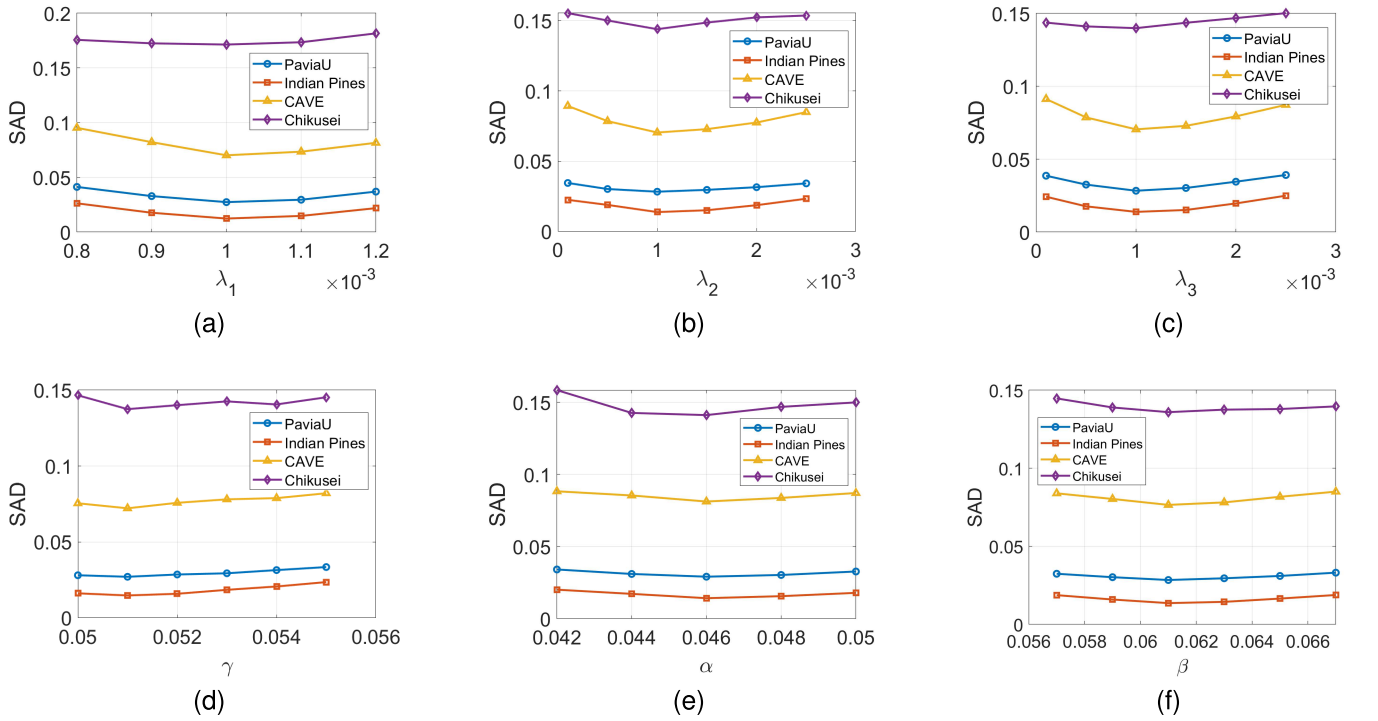


Fig. 9. Parameter sensitivity analysis and super-resolution results on various datasets. (a) Effect of λ_1 on SAD. (b) Effect of λ_2 on SAD. (c) Effect of λ_3 on SAD. (d) Effect of γ on SAD. (e) Effect of α on SAD. (f) Effect of β on SAD.

it obtains the highest PSNR of 32.0589, surpassing the second-best method (SCDM) by a large margin of more than 3.2 dB, indicating superior fidelity in the reconstructed HSI. Furthermore, our method reports the lowest RMSE (0.0199) and SAD (0.1357), reflecting a more accurate spectral restoration and fewer spectral distortions. Notably, it also achieves the best ERGAS score (6.7582), which further confirms its effectiveness in preserving global spectral consistency. Although SCDM shows competitive results among deep learning-based methods, its ERGAS (9.6697) and SSIM (0.7669) are still inferior to ours. Compared to all other baselines, including classical methods such as J-SLoL, DTS, and SSIC, our model consistently outperforms them by a significant margin. These results clearly demonstrate the effectiveness of our method in producing high-quality and spectrally consistent results. Fig. 8 presents a qualitative comparison of the reconstructed HSIs on the 25th spectral band of the Chikusei dataset. Evidently, our proposed method yields reconstructions with finer structural details and significantly lower residual errors, underscoring its superior reconstruction fidelity. In addition, Fig. 5(d) illustrates the spectral profiles averaged over a 5×5 pixel region. The spectral curve obtained by our method aligns almost perfectly with the ground truth, validating its effectiveness in preserving spectral integrity and capturing subtle spectral variations.

C. Parameters Analyses

As illustrated in (10), the proposed model involves three key hyperparameters, namely, λ_1, λ_2, and λ_3, which play a crucial role in governing its overall performance. In this section, we discuss how to select them in the four test datasets as shown in Fig. 9. Specifically, the sparse regularization parameter λ_1

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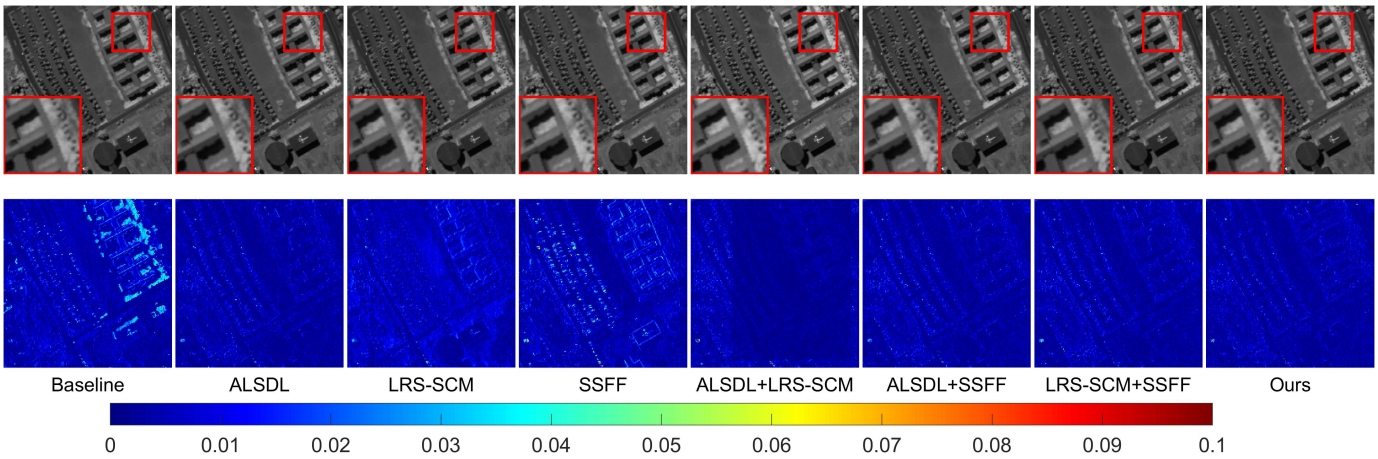


Fig. 10. Visual comparisons of ablation variants on the PaviaU. Results are shown for the 80th spectral band. The top row illustrates the zoomed-in view reconstructed outputs, and the bottom row shows the associated pseudo-colored residuals, highlighting differences across model configurations.

was optimally set to 0.001. A lower value led to insufficient sparsity, whereas a higher value caused excessive information loss [see Fig. 9(a)]. The spectral correlation regularization parameter λ_2 was also set to 0.001, resulting in a minimum SAD of 0.283 [see Fig. 9(b)]. A lower value retained excessive redundancy, while a higher value excessively smoothed the spectral features. The spatial consistency constraint parameter λ_3 was set to 0.001, which balanced spatial structure and spectral fidelity, reducing the SAD to 0.282 [see Fig. 9(c)].

Beyond the primary model analysis, we further examined three key parameters drawn from two auxiliary models [see (6) and (8)] to investigate their influence on the overall reconstruction performance. Specifically, the low-rank regularization coefficient γ was set to 0.051, effectively balancing the preservation of fine spectral details with the suppression of spectral redundancy, and yielding the lowest SAD of 0.271 [see Fig. 9(d)]. In addition, two essential parameters used to construct the spectral correlation weight matrix W_i, namely, α and β, were explored. The configuration $\alpha = 0.046$ and $\beta = 0.061$ achieved the optimal SAD, further confirming the efficacy of the proposed correlation modeling strategy [see Fig. 9(e) and (f)].

In addition to these, several secondary parameters were empirically determined. The number of clusters in the K-means algorithm was set to 320. The sparsity regularization parameter for the coefficient matrix was set at 0.001, and the number of dictionary atoms was set at 28.

It should be noted that the same parameter configuration, when transferred to the Indian Pines, CAVE, and Chikusei datasets, consistently yielded stable and high-quality super-resolution performance (see Fig. 9), further validating the generalizability and robustness of the proposed method.

D. Ablation Study and Analysis

In the ablation study, we discuss the importance of three models, including ALSDL, LRS-SCM, and SSFF. Since ALSDL applies the grouping strategy to learn local spectral dictionaries, we define a baseline method, which learns the global spectral from the auxiliary HSI, to test the effectiveness of ALSDL. In specific, we first learn a global hyperspectral dictionary $\mathbf{D}_G$ from the auxiliary HSI $\mathbf{Z}$ by using

$$\arg\min_{\mathbf{D}_G} \|\mathbf{Z} - \mathbf{D}_G\mathbf{A}_G\|_F^2 + \gamma_1 \|\mathbf{D}_G\|_* + \eta_1 \|\mathbf{A}_G\|_1 . \tag{36}$$

The hyperspectral dictionary $\mathbf{D}_G \in \mathbb{R}^{\lambda_H \times L_G}$ is used to solve the sparse coefficients of the observed HR-MSI $\mathbf{Y}$ using

$$\arg\min_{A} \|\mathbf{Y} - \mathbf{F}\mathbf{D}_G\mathbf{A}\|_F^2 + \eta_2 \|\mathbf{A}\|_1 . \tag{37}$$

The desired HR-HSI is obtained by $\mathbf{X} = \mathbf{D}_G\mathbf{A}$.

To evaluate the contributions of each proposed module, we conduct a comprehensive ablation study on the PaviaU. Quantitative results are reported in Table V, while qualitative visualizations are shown in Fig. 10.

Compared to the baseline method, integrating the ALSDL module alone improves PSNR by 5.02 dB and reduces SAD by 60.8%, confirming its effectiveness in enhancing spectral representation. This gain is attributed to its ability to partition spectral subspaces via K-means clustering and to construct localized dictionaries that take advantage of the full spectral content of the auxiliary HSI-addressing the limitations of global dictionaries in heterogeneous scenes. When combined with LRS-SCM, the model achieves further gains: PSNR increases to 45.60 dB, SSIM rises to 0.9894, and ERGAS drops to 1.2885. These improvements demonstrate that the low-rank sparse constraint effectively models interband correlations and suppresses spectral redundancy. Alternatively, the combination of ALSDL and SSFF achieves even lower SAD (0.0271) and RMSE (0.0058), alongside a higher SSIM (0.9936), indicating that the spectral similarity filtering mechanism (SSFF) effectively mitigates spatial misalignments, particularly in texture-rich regions. Integrating all three components leads to the best performance, with PSNR reaching 46.27 dB and ERGAS further reduced to 1.2527. These results confirm the complementary nature of the proposed modules: ALSDL provides localized spectral representations, LRS-SCM enhances spectral smoothness through low-rank regularization, and SSFF enforces spatial consistency through nearest-neighbor spectral matching.

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TABLE V
ABLATION STUDY RESULTS

Method	PSNR	SAD	RMSE	SSIM	ERGAS
Baseline Method	40.0233	0.0774	0.0131	0.9787	3.2058
ALSDL	45.0418	0.0313	0.0066	0.9889	1.4632
LRS-SCM	43.8693	0.0385	0.0080	0.9900	1.7350
SSFF	42.7440	0.0380	0.0085	0.9568	1.9067
ALSDL + LRS-SCM	45.6041	0.0297	0.0062	0.9894	1.2885
ALSDL + SSFF	45.4152	0.0271	0.0058	0.9936	1.3255
LRS-SCM + SSFF	44.9711	0.0315	0.0066	0.9923	1.4660
Ours	**46.2673**	**0.0263**	**0.0056**	**0.9943**	**1.2527**
Ideal Value	$\uparrow \infty$	$\downarrow 0$	$\downarrow 0$	$\uparrow 1$	$\downarrow 0$

Fig. 10 further validates these findings with visual comparisons of the 80th spectral band. The baseline reconstruction exhibits noticeable artifacts and blurred structures. Introducing ALSDL enhances edge sharpness and reduces residuals. Adding LRS-SCM yields smoother spectral transitions, while substituting it with SSFF sharpens textures and suppresses residuals in fine-detail regions. The complete model produces the most faithful reconstruction, with minimal errors across both spectral and spatial dimensions.

E. Complexity Analysis

The proposed algorithm consists of three main components: ALSDL, LRS-SCM, and SSFF. We analyze the time complexity of each part individually. In the ALSDL stage, the K-means clustering process has a complexity of $O(NPT)$, where N is the total number of pixels, P is the number of clusters, and T is the number of iterations. Spectral downsampling involves a dimensionality transformation matrix, with a complexity of $O(\lambda_M \lambda_H N_Z)$, where λ_M and λ_H represent the spectral dimensions of MSI and HSI, respectively, and N_Z is the number of spatial pixels in the auxiliary HSI. Spectral dictionary learning, performed through alternating optimization, has a complexity of $O(L^2\lambda_H + L\lambda_H^2)$, where L denotes the number of dictionary atoms. In the LRS-SCM stage, the construction of the spectral correlation matrix involves matrix transposition and low-rank sparse constraints, with a complexity of $O(P\lambda_H^2 + \lambda_H^3)$. In the SSFF stage, constructing the reconstruction term has a complexity of $O(\lambda_H N)$. The final optimization process for the data fidelity term contributes $O(\lambda_M LN)$, while the spectral correlation term contributes $O(\lambda_H^2 N)$. Under the assumption that $\lambda_M \ll \lambda_H$, the overall time complexity is dominated by the dictionary atom count k and the spectral dimension λ_H, and is expressed as $O(PNT + L^2N + L\lambda_H^2 + \lambda_H^2 N)$.

Table VI reports the runtime comparisons across four benchmark datasets. On the PaviaU, our method completes in 327.25 s, performing competitively with UPSL_SSR (231.44 s), DTS (367.49 s), and SSIC (284.96 s), while remaining reasonably close to SLA_DOL (239.49 s) and J-SLoL (213.58 s), the fastest baseline. For the Indian Pines dataset, our method achieves a runtime of 161.46 s, ranking third behind SLA_DOL (139.70 s) and SCDN (151.55 s), and outperforming methods like SSRAN (171.58 s), UPSL_SSR (171.19 s), and DTS (185.65 s). On the CAVE dataset, which features greater spectral complexity, our approach completes in 326.52 s, clearly outperforming deep learning-based competitors such as MSFN (375.44 s), SSRAN (427.89 s), and LTRN (361.72 s), and offering a strong tradeoff between accuracy and efficiency despite J-SLoL (134.31 s) reporting the lowest runtime due to its simplified pipeline. On the Chikusei dataset, our method runs in 347.53 s, showing consistent efficiency relative to most deep methods, including MSFN (426.75 s) and SCDN (396.64 s), and only marginally slower than UPSL_SSR (179.29 s) and SLA_DOL (135.74 s). In summary, while some lightweight models such as J-SLoL achieve faster runtimes, our method consistently balances computational cost and reconstruction performance across diverse scenes, demonstrating particular robustness on more challenging datasets such as CAVE and Chikusei.

TABLE VI
COMPARISON OF RUNNING TIMES OF DIFFERENT METHODS (SECONDS)

Method	PaviaU	Indian Pines	CAVE	Chikusei
J-SLoL [29]	**213.58**	234.25	**134.31**	194.60
DTS [28]	367.49	185.65	248.50	204.33
UPSL_SSR [30]	231.44	171.19	291.62	179.29
SSIC [50]	284.96	246.35	235.61	227.10
SLA_DOL [40]	239.49	**139.70**	570.18	**135.74**
MSFN [51]	368.87	242.37	375.44	426.75
SSRAN [52]	319.23	171.58	427.89	378.98
LTRN [53]	358.74	189.20	361.72	385.20
SCDN [54]	337.51	151.55	377.91	396.64
Ours	327.25	161.46	326.52	347.53

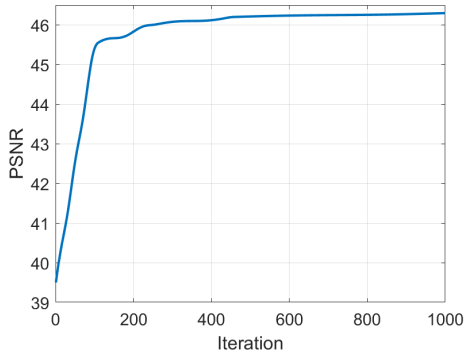


Fig. 11. Convergence analysis.

F. Convergence Analysis

As illustrated in Fig. 11, the PSNR rapidly increases within the first 100 iterations and gradually converges to around

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46.3 dB. This shows the ability of the model to quickly capture global spectral structures and progressively refine local details. The convergence behavior reflects the stability of the optimization framework in (10), which integrates spectral fidelity, sparse representation, low-rank modeling, and spectral matching. Despite the nonconvexity introduced by sparsity, the alternating optimization scheme ensures a monotonic decrease in the objective, leading to convergence to a local minimum.

VI. Conclusion

This article presented AHSIG-SSR, a framework for the SSR task, which addresses the limitations of the traditional registration-dependent, prior-dependent, spectral library-dependent, and massive training sample-dependent methods. The proposed AHSIG-SSR not only improves the SSR accuracy but also has a simple operation. In specific, we designed three different models, including ALSDL, LRS-SCM, and SSFF, to capture the spectral information from only an auxiliary HSI, which has spectral information similar to that of the desired HSI. Extensive experiments on benchmark datasets, including Pavia University, Indian Pines, and CAVE, demonstrate that the proposed method not only achieves superior quantitative performance (e.g., PSNR of 46.27 dB and SSIM of 0.9943 on Pavia University), but also exhibits strong cross-dataset generalization without the need for pretraining or registered data. These results highlight the robustness, flexibility, and practical applicability of our approach in real-world multimodal imaging scenarios. Future work will explore the integration of deep neural networks to further capture nonlinear spectral–spatial relationships and enhance model adaptability, as well as the expansion of spectral prior knowledge through comprehensive library construction. Overall, this study provides an effective and generalizable solution for SSR in unregistered imaging systems, with broad potential in remote sensing, environmental monitoring, and medical diagnostics.

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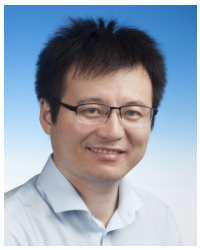
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